Reflections on compressed sensing

Terence Tao · 25 May 2009

https://terrytao.wordpress.com/2009/05/25/reflections-on-compressed-sensing/

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[This post should have appeared several months ago, but I didn't have a link to the newsletter at the time, and I subsequently forgot about it until now. -T.]

Last year, Emmanuel Candés and I were two of the recipients of the 2008 IEEE Information Theory Society Paper Award, for our paper "Near-optimal signal recovery from random projections: universal encoding strategies?" published in IEEE Inf. Thy.. (The other recipient is David Donoho, for the closely related paper "Compressed sensing" in the same journal.) These papers helped initiate the modern subject of *compressed sensing*, which I have talked about earlier on this blog, although of course they also built upon a number of important precursor results in signal recovery, high-dimensional geometry, Fourier analysis, linear programming, and probability. As part of our response to this award, Emmanuel and I wrote a short piece commenting on these developments, entitled "Reflections on compressed sensing", which appears in the Dec 2008 issue of the IEEE Information Theory newsletter. In it we place our results in the context of these precursor results, and also mention some of the many active directions (theoretical, numerical, and applied) that compressed sensing is now developing in.

Comments (11)

 **Igor Carron** · 28 May 2009

Dear Terry,
Thanks to you and Emmanuel for the plug about the blog in the newsletter!
Igor.
http://nuit-blanche.blogspot.com

 **weiyu** · 29 July 2009

just one simple question from the paper "Near Optimal Signal Recovery from Random Projections:Universal Encoding Strategies?"
For the Fourier ensemble, in order for the R.I.P conditions to hold, can we get the sparsity level, namely the cardinality of the signal support, be linearly growing with the problem dimension N, if we allow the number of freuqnecy sample grow linearly with the problem dimension N? In the paper, we can have N/log(N) growth, can we get rid of this constraint?
Many thanks for the reply.

 **Terence Tao** · 3 August 2009

In the Gaussian case, this can be achieved; see for instance my joint paper with Candes, Rudelson, and Vershynin at http://www.math.ucla.edu/~tao/preprints/FOCS05.pdf (though the observation probably predates the explicit introduction of the RIP concept).

 **Anonymous** · 11 July 2011

In reading the proof of the UUP for the Fourier matrix from the "Universal Encoding Strategy paper", http://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=4016283
two questions:
1. In the definition of the set {J_0,...,J_1} (right above Eq. (7.48) in the IEEE version), should the N^2 in the definition be N^{-2}?
2. In the telescoping expansion below Eq. (7.48). Is the expansion written in the right way. In this expansion, f_{J_{0}} was subtracted twice. Was there a typo there?
Thank you very much for your reply, which will greatly help understanding of this paper.

 **Terence Tao** · 14 July 2011

Yes, there should be a minus sign in both cases (N^2 should be N^{-2}, and the telescoping series should have its sign reversed.)

 **Richard** · 22 November 2011

Dear Terry,
In your paper "Decoding by Linear Programming" you wish to recover an input vector $f \in \mathbf{R}^n$ from corrupted measurements $y = Af + e$. Can one change f to be a square matrix? In other words, could you also ask given corrupted measurements can you recover an input matrix?

 **Anonymous** · 26 September 2012

Recently, I published a paper on the super robustness at:
http://arxiv.org/abs/1208.1810
Is there any insight on the relation between our approach and GGD used in compressive sensing? The main goal of our approach was to do pattern matching.
Any suggestions to extend the results to more general transformation than translation estimation?

 **Anonymous** · 27 September 2012

Sorry for the typo: “GGD used in compressive sensing” should read “GGD=based approaches used in compressive sensing”.

 **Anonymous** · 29 September 2012

Mathematically, compressive sensing and super robustness of location estimation studied in http://arxiv.org/abs/1208.1810 are all related to a linear system $A=BC$. The difference is that in compressive sensing, A and B are all reliable but there are not enough equations to solve C, thus a “good guess” is generated using linear programming based on some kind of mathematics assumption; in the super robustness of location estimation, A and B contains a lot of noise or majority of the data in A and B are not reliable, the paper presents an approach how to generate a reliable C from the very noisy data.

 **Anonymous** · 6 September 2013

The newsletter is missing from the website. Could Professor Tao help upload a direct link to the newsletter article here? Thanks.

 **Igor Carron** · 26 September 2013

Anonymous,
It is here: http://www-stat.stanford.edu/~candes/papers/itNL1208.pdf
Igor.